

Ankit Raj

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SUMMARY

Eager data science researcher with programming (Python, R) and machine learning skills (Random Forest, data cleaning). Seeks to leverage academic knowledge and research for real-world impact.

WORK EXPERIENCE

Machine Learning and GEN AI Researcher

July 2025 - Present

Guide : Prof. Alankar Alankar

- AI Based Condition Monitoring, Prediction Of Health and Remaining life of components of Power Electronics.

Data Analyst(Sparks Foundation)

Dec 2022 - Feb 2023

- Streamlined data collection and reporting procedures, reducing processing time by 20% enhancing efficiency.
- Implemented process improvements and automation solutions, resulting in 15% increase in productivity.
- Creating visualizations to convey stories using charts and graphs. Skills used - EDA,EDA tools, Python, Data Visualization.

Web Developer(IIT BOMBAY)

Jan 2022 - March 2022

- Updated WordPress themes and plugins on professor's web page project by adding 20+ new features using PHP, JavaScript, HTML, and CSS.
- Designed and implemented a responsive e-commerce platform, increasing mobile traffic by 50% and achieving a 30% boost in overall sales conversion rates using HTML5, CSS3, JavaScript, and React.
- Added a countdown timer and 10+ features to the company's website for an upcoming product preview.

PROJECTS

Breast Cancer Detection

Achieves high accuracy (around 95%) by combining multiple decision trees, making it vigorous to over-fitting.Deliver unprecedented accuracy (potentially exceeding 95%) by iteratively building decision trees, focusing on areas with prior model errors .

Forest Fire Prediction

Machine learning algorithms such as Random Forest or LSTMs (Long Short-Term Memory networks) analyze this data to identify patterns. These models can achieve accuracy exceeding 90% in identifying areas with a high risk of fire. This model leverages machine learning algorithms to analyze historical data and predict the likelihood of forest fires less than 1 secs.

Heart Disease Detection

Offers a pretty baseline (around 80% accuracy) and interpretability of results, making it useful for understanding which factors most influence predictions.Can potentially reach even higher accuracy (up to 95%) by focusing on areas where previous models struggle, leading to potentially more precise predictions.

EDUCATION

2025 - present	PhD (Machine Learning and Data Science) at IIT BOMBAY	
2021 - 2025	Bachelor's Degree at Quantum University , Roorkee	GPA: 8.35
2021	Class 12th CBSE Board	91%
2019	Class 10th CBSE Board	81%

SKILLS

Languages:	Python , MySQL , C++
Frameworks:	Pandas , Numpy , Scikit - Learn , Matplotlib , Flask
Tools:	Power BI , Excel , PowerPoint , Tableau , MySQL
Platforms:	PyCharm , Jupyter Notebook , Visual Studio Code , Spyder
Certifications:	Data Analysis Using Python , Machine Learning Using Python